

New Eden Digital Twin

Reference Sources & Standards

This document lists the real-world machines, robots, construction methods, and engineering standards that the New Eden browser-based digital twin (digital-twin/index.html) was designed with reference to. The digital twin is a conceptual visualization and interactive proof-of-concept — it is **not** a certified engineering deliverable, and nothing in it should be read as a claim of compliance with any standard listed below. See the disclaimer at the end of this document.

1. Roll-Forming Production Line

The digital twin's roll-forming station (decoiler → servo feeder → 20-stand roll former → hydraulic cutoff → outfeed) is modeled in the style of, and dimensioned with reference to, steel floor-deck / joist roll-forming machines published by Wuxi Sunway Machinery Co., Ltd. An exact public specification page for a model literally numbered “SW-FJ300” could not be independently re-verified at the time this document was compiled; the layout and station count instead reflect Sunway's published floor-deck roll-forming machine product category generally, which shares the same uncoiler / leveller / roll-form / punch / shear / stacker architecture.

swforming.com — SUNWAY Machine, manufacturer homepage

[Floor Deck Roll Forming Machine](#) — product page

[Floor Deck Roll Forming Machine](#) — product category

2. Robotic Welding Cell — ABB IRB 8700-550/4.20

The truss-welding robot cell is a spec-accurate reconstruction of the ABB IRB 8700-550/4.20, ABB's largest 6-axis industrial robot, based on ABB's own published product and datasheet pages. Key published specifications used: 550 kg payload at 4,200 mm reach (620 kg wrist-down / 475 kg with LeanID); single motor and gear per axis; mechanical-spring counterbalance (no gas springs); IP 67 / Foundry Plus 2 protection rating; ~725 kgm² moment of inertia; 200–600 V supply at ~3.93 kW; RoHS 2002/95/EC and REACH No 1907/2006 conformance as stated by ABB.

one.robotics.abb.com — IRB 8700 official product page

abb.com — IRB 8700, Large Robots

[ABB — IRB 8700 official datasheet \(PDF\)](#)

[ABB — IRB 8700 / IRC5 Product Specification \(PDF, doc 3HAC052852\)](#)

3. Automated Steel-Frame Wall & Floor Framing Tables

The Walls lane's step-by-step build sequence (bottom track, top track, king studs, field studs picked one at a time from a magazine, window opening framing, diagonal strap bracing, fastening pass) reflects the general operating principle of automated cold-formed-steel framing tables used across the light-gauge steel construction industry — the category the New Eden master vision document references (in the spirit of Autovol-style automated framing factories). These machines typically roll-form coil steel into C-shaped studs and tracks to AISI S240 profiles, then hold members square on a table for automated or semi-automated fastening, punching connection holes and labelling each piece for assembly.

BuildSteel.org — Robotic Automation in Cold-Formed Steel Construction

[FRAMECAD](#) — automated steel framing systems

[AAC Steel](#) — Precision Manufacturing for Cold-Formed Steel Framing

4. Relevant Engineering & Safety Standards

The standards below are the ones a real, physical version of this factory and its products would need to be engineered against by licensed professionals. They are listed here for reference only. **The digital twin does not claim, and cannot claim, compliance with any of them** — it is a browser visualization, not a certified structural, mechanical, or safety engineering document.

AS/NZS 4600:2018

Design of cold-formed (light-gauge) steel structural members used for load-carrying purposes in buildings — the studs, tracks and joists this factory would roll-form.

<https://www.steel.org.au/what-we-do/focus-areas/cold-formed-light-gauge-steel/standards-and-design/>

AS 4100:2020

Design, fabrication, erection and modification of structural steelwork; referenced by the NCC as the mandatory steel-structures standard for covered projects.

<https://www.steel.org.au/resources/elibrary/resources/tn005-guidelines-for-designing-to-as-4100-when-imp/>

AS/NZS 1170 series (0–4)

Structural design actions — general principles, permanent/imposed loads, wind, snow/ice, and earthquake actions used to size any structure built from this factory's modules.

<https://www.studocu.com/en-au/document/university-of-new-south-wales/concrete-structures/1170-australian-standards/7590289>

AS/NZS 4024.1 series

Safety of machinery — the family of standards (general principles, risk assessment, guarding, e-stops, control systems) that would govern real roll-forming and robotic welding equipment on a factory floor.

<https://www.assessor.com.au/resources/guides/australian-standards-4024-series>

ISO 10218-1:2025 / -2:2025 (AS ISO 10218)

Safety requirements for industrial robots (Part 1, manufacturers) and robot cells / system integration (Part 2) — directly relevant to the IRB 8700 welding cell.

<https://www.iso.org/standard/73933.html>

National Construction Code (NCC)

Australia's uniform building code; the ABCB's dedicated Prefabricated, Modular and Offsite Construction initiative and handbook govern how modular homes like New Eden's are certified for use.

<https://ncc.abcb.gov.au/initiatives/prefabricated-modular-and-offsite-construction>

Disclaimer

This document and the New Eden digital twin it accompanies are conceptual / illustrative materials produced to visualize a proposed autonomous modular home factory. Machine and robot dimensions are grounded in the manufacturers' own published specifications wherever possible, and the standards listed above are the ones a real build would need to be assessed against — but no AI system, and no document it produces, can certify real-world engineering or regulatory compliance. Before any physical construction, procurement, or operation based on this concept, New Eden would need formal review and sign-off from licensed structural, mechanical, electrical, and safety engineers, and compliance verification against the current editions of the applicable standards at the time of build.